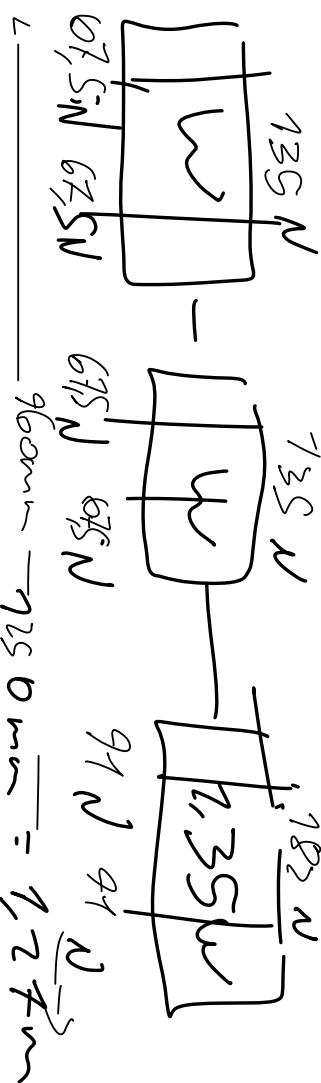
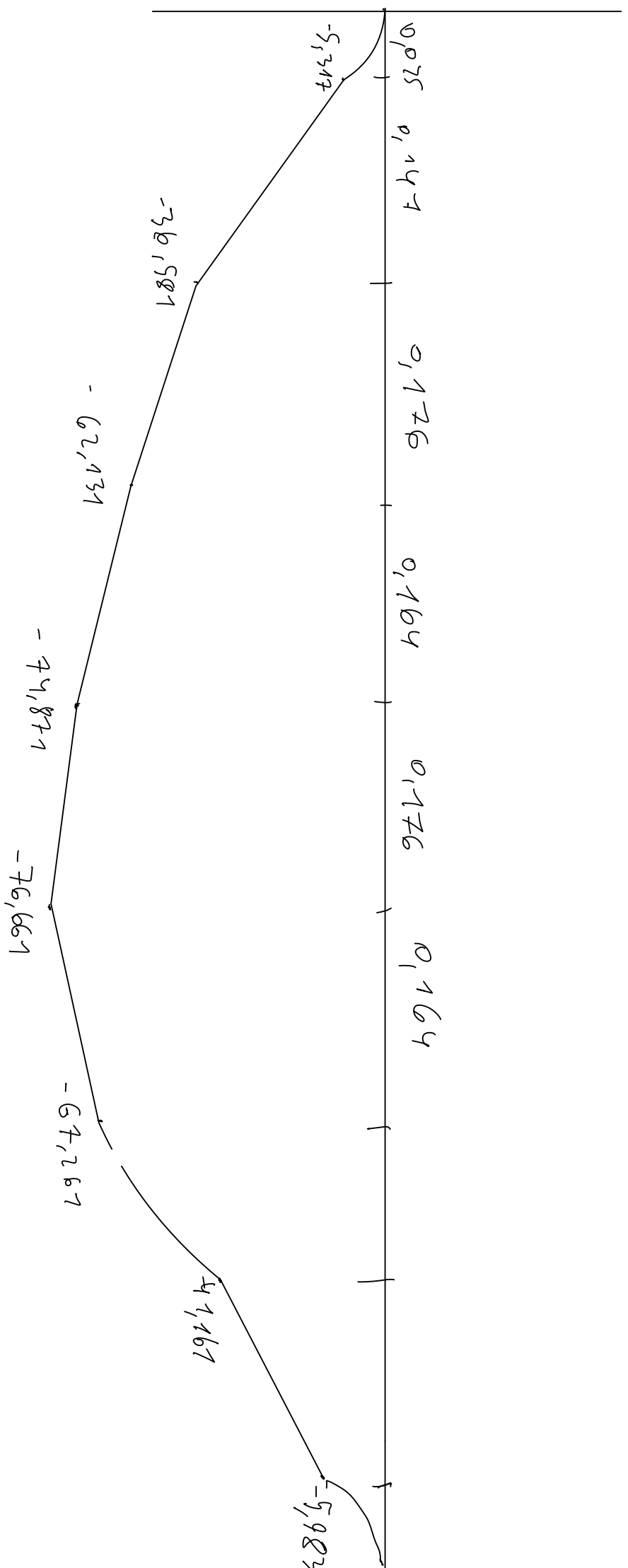
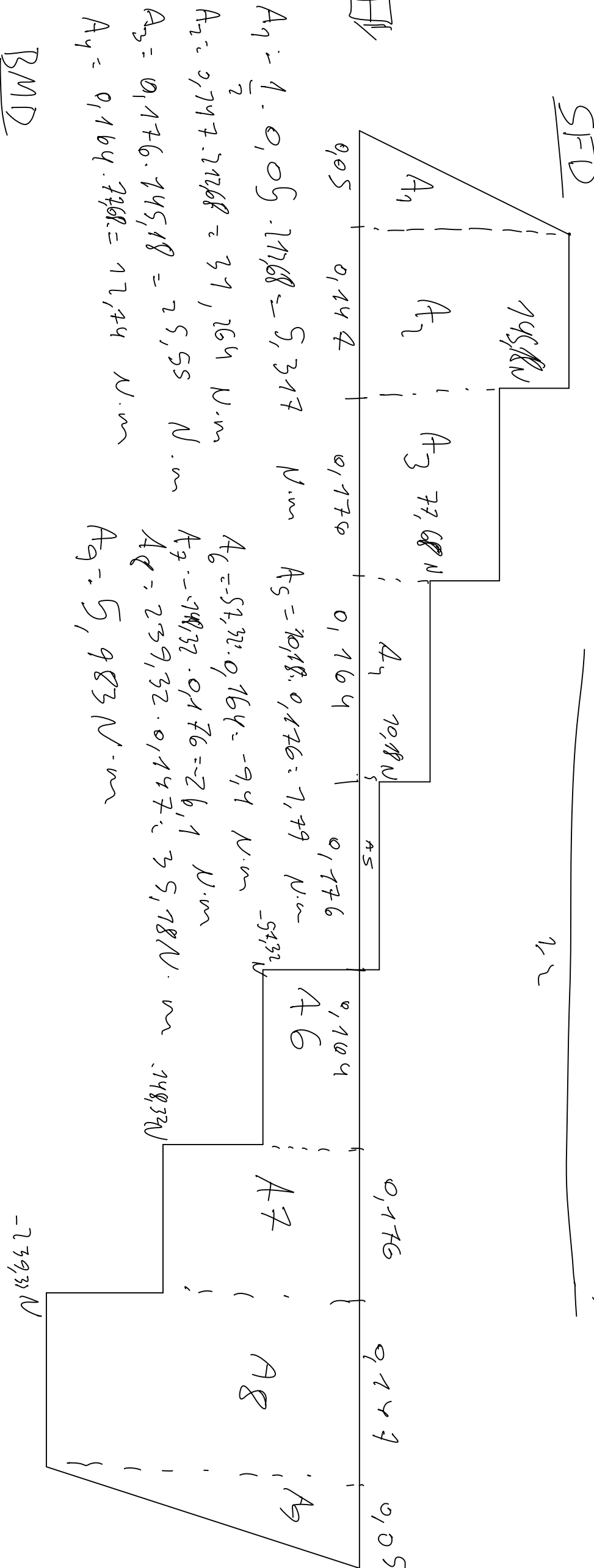
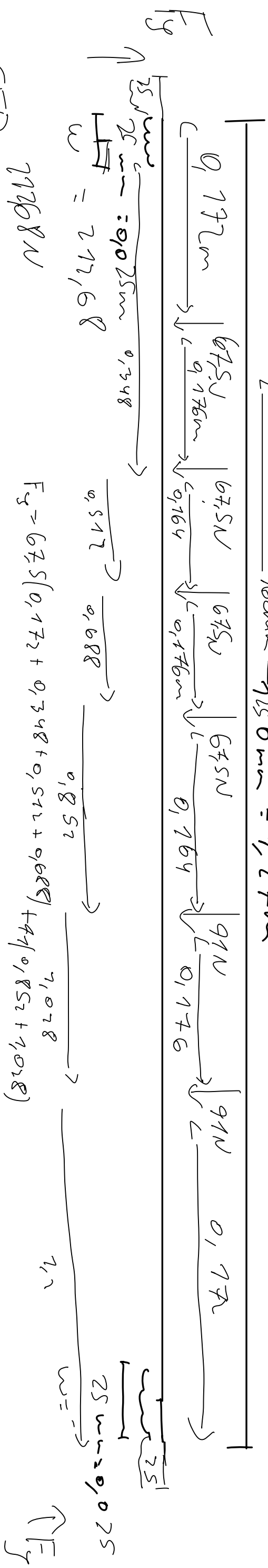


FBD



Total weight = 452 N

$$F_y = 239.32 \text{ N}$$



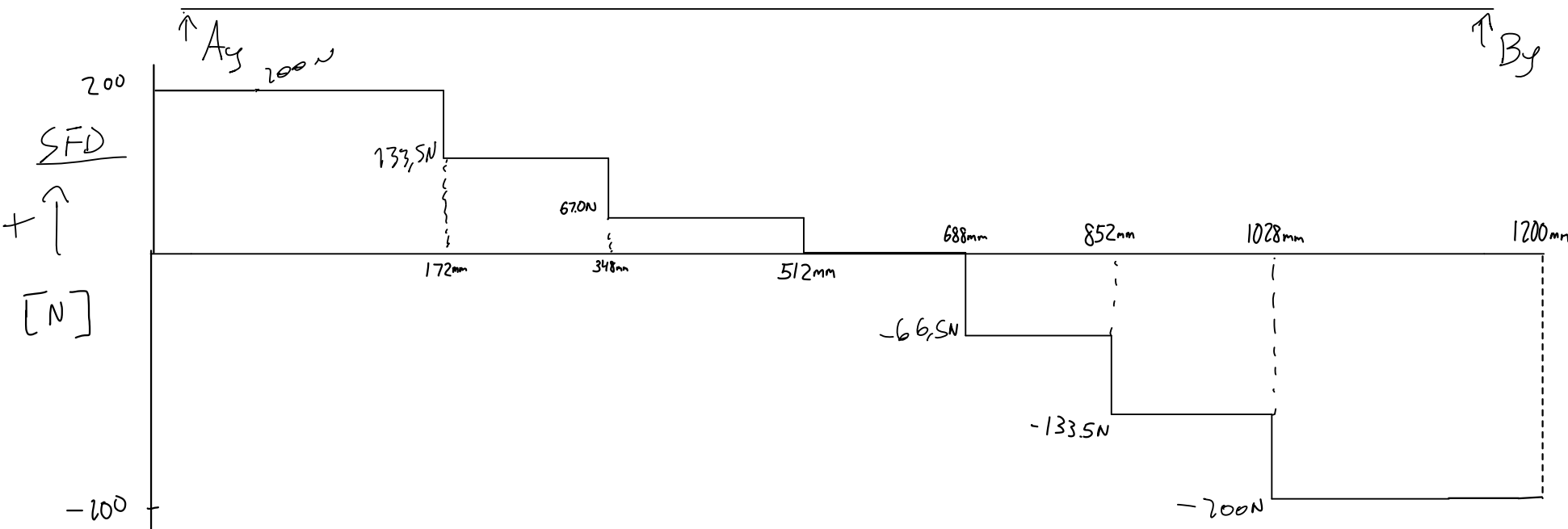
Design Calculations | Load Case 1

SFD and BMD for train at midpoint, but will use max shear force and max bending moment

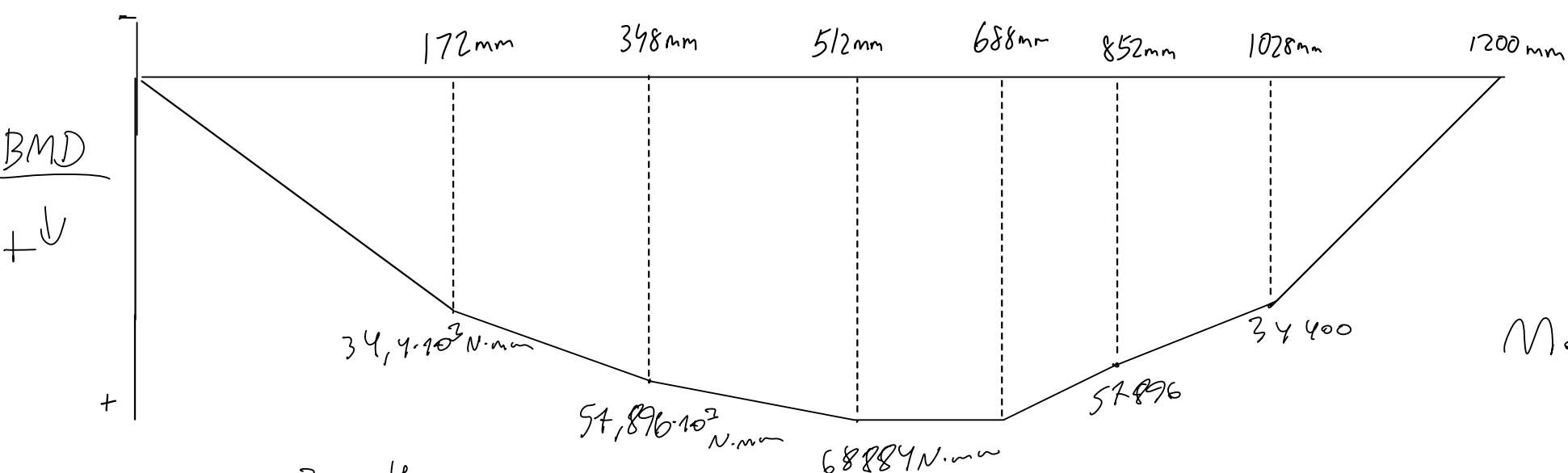
$$\sum M_y = 0$$

$$B_y = \frac{66,5(172 + 348 + 512 + 688 + 852 + 1028)}{1200}$$

$$B_y = A_y = 200 \text{ N}$$



$$\text{Max Shear} = 256,7 \text{ N}$$



$$\text{Max Bending} = 69211,7 \text{ N·mm}$$

$$I = 418 \cdot 10^3 \text{ mm}^4$$

$$\bar{y} = 41,4 \text{ mm}$$

Calculations

1) Compression and stress

$$\text{F.O.S. tension} = \frac{\sigma_{\text{tens}}}{\sigma_{\text{bottom}}} = \frac{30}{6,86} = 4,4$$

$$\text{F.O.S. compression} = \frac{\sigma_{\text{compression}}}{\sigma_{\text{top}}} = \frac{6}{5,78} = 1,04$$

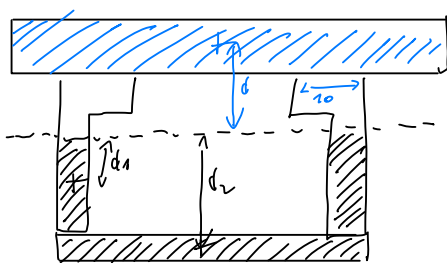
$$\sigma_{\text{bottom}} = \frac{M_{y_{\text{bot}}}}{I} = \frac{69211 \cdot 41,4}{418 \cdot 10^3} = 6,86$$

$$y_{\text{bot}} = 41,4$$

$$\sigma_{\text{top}} = \frac{M_{y_{\text{top}}}}{I} = \frac{68884 \cdot 34,87}{418 \cdot 10^3} = 5,78$$

$$y_{\text{top}} = 76,27 - 41,4 = 34,87$$

2) Shear stresses



$$Q_{\text{glue}} = 100 \cdot 127 \cdot (75,635 - 41,4) = 4347,845$$

$$Q = \sum A_i d_i$$

$$Q_{\text{max at } \bar{y}} = 2 \cdot 127 \cdot 2040 + 80 \cdot 127 \cdot 49765 = 6117,764$$

$$d_1 = \frac{41,4 - 127}{2} = 20$$

$$d_2 = 41,4 - \frac{127}{2} = 40,765$$

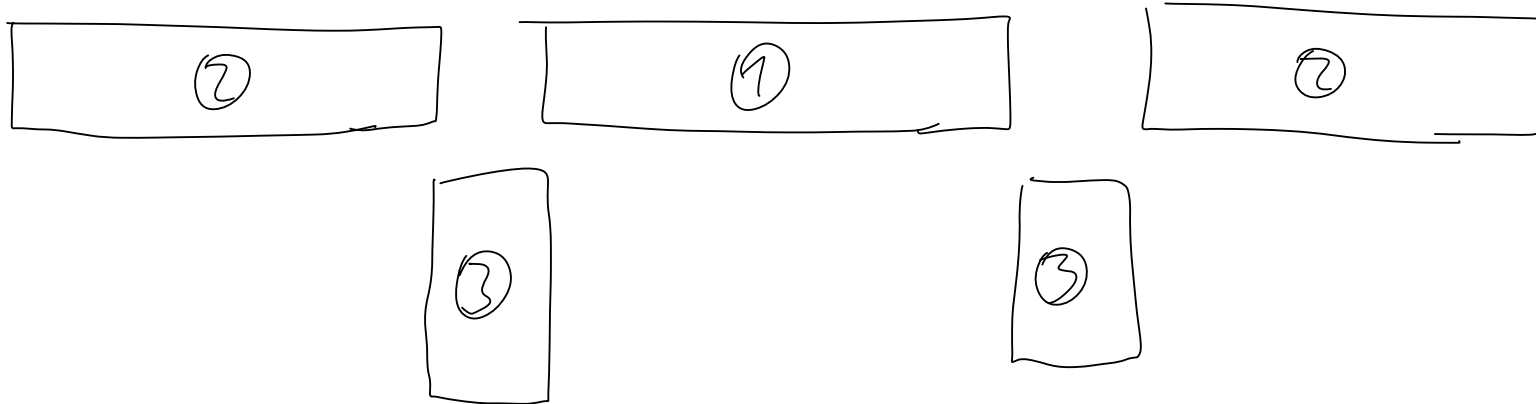
$$\tau_{xy_{\text{max}}} = \frac{V Q_{\text{max}}}{I b} = \frac{751 \cdot 6117,764}{418 \cdot 10^3 \cdot 2 \cdot 127} = 1,48$$

$$\tau_{xy_{\text{glue}}} = \frac{V Q_{\text{glue}}}{I b} = \frac{200 \cdot 4347,842}{418 \cdot 10^3 \cdot 2 \cdot 10} = 0,1337$$

$$\text{F.O.S. glue} = \frac{2}{0,1337} = 15$$

$$\text{F.O.S. shear} = \frac{4}{1,48} = 2,7$$

3) Plate buckling



$$\text{Type 1: } \sigma = \frac{4\pi^2 E}{12(1-\nu^2)} \left(\frac{t}{b} \right)^2 = \frac{4\pi^2 4000}{12(1-0,2^2)} \left(\frac{1,27}{80} \right)^2 = 3,45 \quad \text{F.O.S.} = \frac{3,45}{5,78} = 0,6$$

$$t = 1,27$$

$$b = 77,46$$

$$\text{Type 2: } \sigma = \frac{0,425\pi^2 \cdot E}{12(1-\nu^2)} \left(\frac{t}{b} \right)^2 = \frac{0,425\pi^2 \cdot 4000}{12(1-0,2^2)} \left(\frac{1,27}{10} \right)^2 = 23,5 \quad \text{F.O.S.} = \frac{23,5}{5,78} = 4$$

$$t = 1,27$$

$$b = 10$$

$$\text{Type 3: } \sigma = \frac{6\pi^2 E}{12(1-\nu^2)} \left(\frac{t}{b} \right)^2 = \frac{6\pi^2 \cdot 4000}{12(1-0,2^2)} \left(\frac{1,27}{33,6} \right)^2 = 29,34 \quad \text{F.O.S.} = \frac{29,34}{5,78} = 5,01$$

$$b = 75-41,4 = 33,6$$

$$t = 1,27$$

$$\text{Type 4: } \sigma = \frac{5\pi^2 E}{12(1-\nu^2)} \left[\left(\frac{t}{h} \right)^2 + \left(\frac{t}{a} \right)^2 \right] = \frac{5\pi^2 \cdot 4000}{12(1-0,2^2)} \left[\left(\frac{1,27}{73,73} \right)^2 + \left(\frac{1,27}{400} \right)^2 \right] = 5,25 \quad \text{F.O.S.} = \frac{5,25}{1,48} = 3,55$$

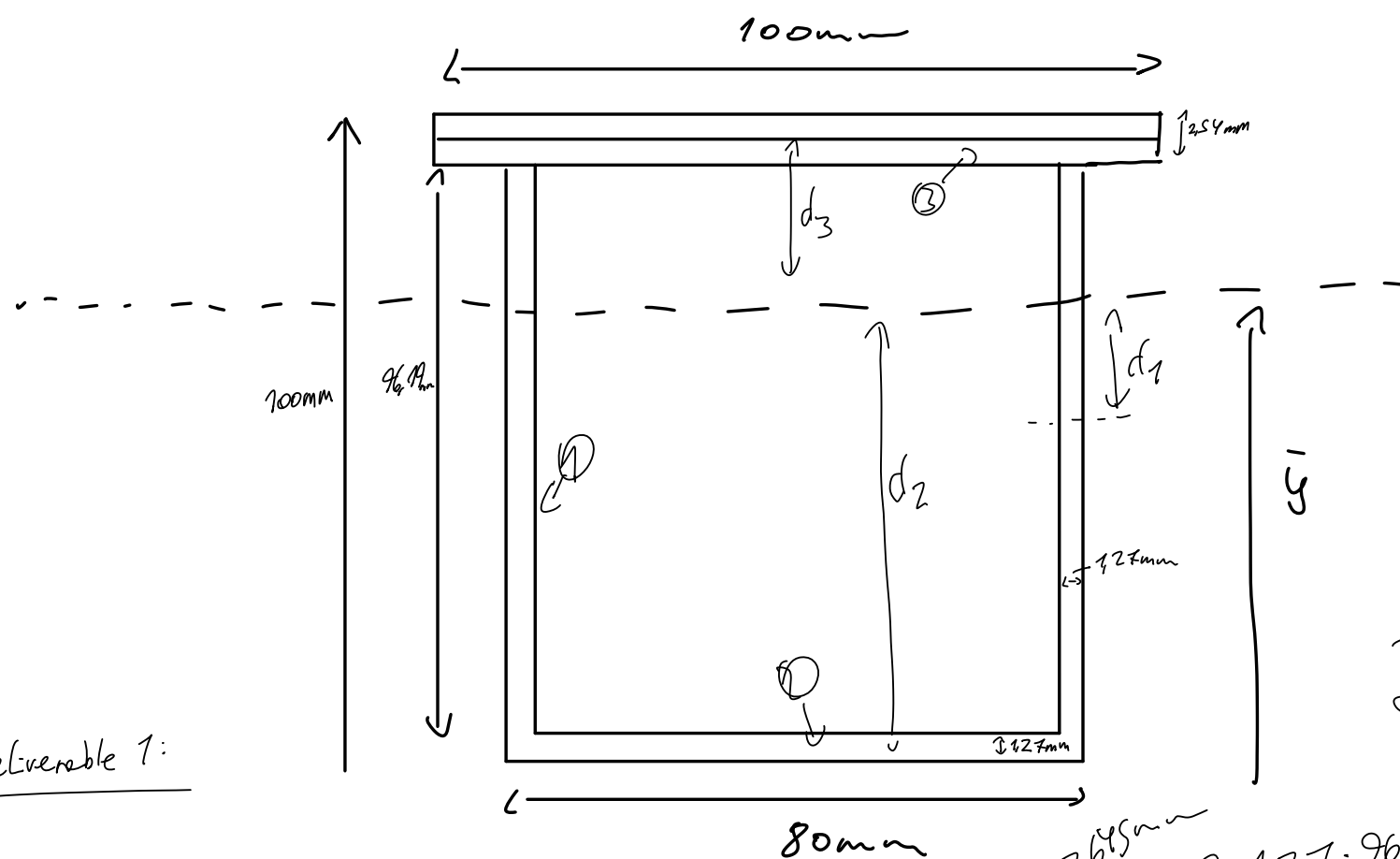
$$h = 73,73$$

$$a = 400$$

$$t = 1,27$$

It will fail in type 1 plate buckling under a load of $0,6 \cdot 400 = 240 \text{ N}$

Final design calculations



$$\bar{y} = \frac{\sum A_i \cdot d_i}{\sum A_i} = \frac{2.54 \cdot 100 \cdot 98.73 + 2 \cdot 96.19 \cdot 49.365 \cdot 1.77}{2.54 + 101.6 + 747.32} = 67.01$$

$$\bar{y} = 67.01$$

$$I = 1.88 \cdot 10^5 + 13.65 + 136.56 + 2.54 \cdot 36.72^2 + 101.6 \cdot 61.375^2 + 12.64 \cdot 244^2 = 9.5 \cdot 10^5 \text{ mm}^4$$

From deliverable 1:

Max bending moment = 78099 N.m

Max shear force = 303.35 N

$$d_1 = 62.01 - 49.365 = 12.645 \text{ mm}$$

$$d_2 = 62.01 - \frac{12.7}{2} = 61.375 \text{ mm}$$

$$d_3 = 100 - 1.77 - 62.01 = 36.72 \text{ mm}$$

$$I_1 = \frac{2 \cdot 12.7 \cdot 96.19^3}{12} = 1.88 \cdot 10^5$$

$$I_2 = \frac{80 \cdot 12.7^3}{12} = 13.65$$

$$I_3 = \frac{100 \cdot 2.54^3}{12} = 136.56$$

① Flexural stress calculations:

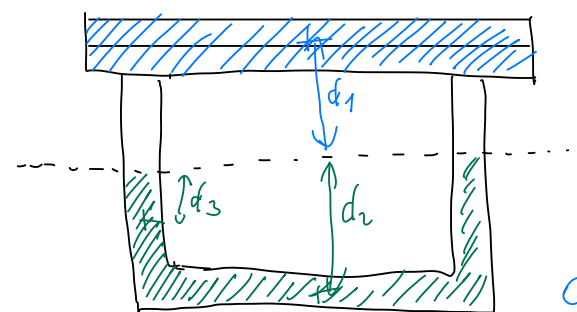
$$\sigma = \frac{M \cdot y}{I} ; \sigma_t = \frac{M \cdot y_{\text{bot}}}{I} = \frac{78099 \cdot 62.01}{9.5 \cdot 10^5} = 5 \text{ MPa} ; \sigma_c = \frac{78099 \cdot 38}{9.5 \cdot 10^5} = 3.1 \text{ MPa}$$

$$y_{\text{bot}} = \bar{y} = 62.01$$

$$y_{\text{top}} = 100 - 62.01 = 38$$

$$\text{F.O.S. tension} = \frac{30}{5} = 6 ; \text{F.O.S. compression} = \frac{6}{3} = 2$$

② Shear stress at glue and centroid



$$Q_{\text{max}} = 2.143 \cdot \left(62.01 - 1.27 \right) \cdot \left(62.01 - \left(\frac{62.01 - 1.27 + 1.27}{2} \right) \right) + 62.01 - \frac{1.27}{2} \cdot 80 \cdot 1.27 = 12618.27$$

$$Q_{\text{glue}} = 100 \cdot 2.54 \cdot \left(100 - 12.7 - 62.01 \right) = 9326.88$$

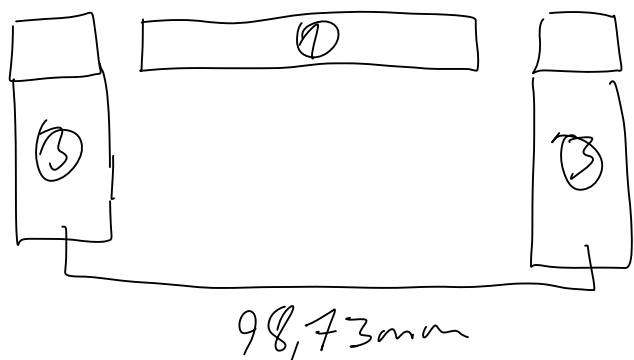
$$\text{F.O.S. shear max} = \frac{4}{1.586} = 2.52$$

$$\text{F.O.S. shear glue} = \frac{2}{0.15} = 13.3$$

$$\tau_{\text{max}} = \frac{V Q_{\text{max}}}{I b} = \frac{739.35 \cdot 12618.27}{9.5 \cdot 10^5 \cdot 2.54} = 1.586$$

$$\tau_{\text{glue}} = \frac{V Q_{\text{glue}}}{I b} = \frac{739.35 \cdot 9326.88}{9.5 \cdot 10^5 \cdot 20} = 0.15$$

③ Plate buckling



$$\text{Case 1: } \sigma = \frac{4\pi^2 E}{12(1-\nu^2)} \left(\frac{t}{b} \right)^2 = \frac{4\pi^2 \cdot 4000}{12(1-0,2^2)} \left(\frac{2,54}{98,73} \right)^2 = 9,07$$

$$FOS_1 = 2,93$$

$$\sigma_1 = 9,07 \text{ MPa}$$

$$\text{Case 3: } \sigma_3 = \frac{6\pi^2 E}{12(1-\nu^2)} \left(\frac{t}{b} \right)^2 = \frac{6\pi^2 \cdot 4000}{12(1-0,2^2)} \left(\frac{1,27}{35,45} \right)^2 = 26,39 \quad FOS_3 = 8,5$$

$$\text{Case 4: } \sigma_4 = \frac{5\pi^2 E}{12(1-\nu^2)} \left[\left(\frac{t}{h} \right)^2 + \left(\frac{t}{b} \right)^2 \right] = \frac{5\pi^2 \cdot 4000}{12(1-0,2^2)} \left[\left(\frac{1,27}{96,19} \right)^2 + \left(\frac{1,27}{100} \right)^2 \right] = 5,75 \quad FOS_4 = 3,64$$

Factors of safety: Failure mode: compression

1. Compression: 1,9

2. Tension: 6

3. Shear: 2,52

4. Glue: 13,3

5. Case 1 buckling: 2,93

6. Case 3 buckling: 8,5

7. Case 4 buckling: 3,64

Code outputs:

Design 0:

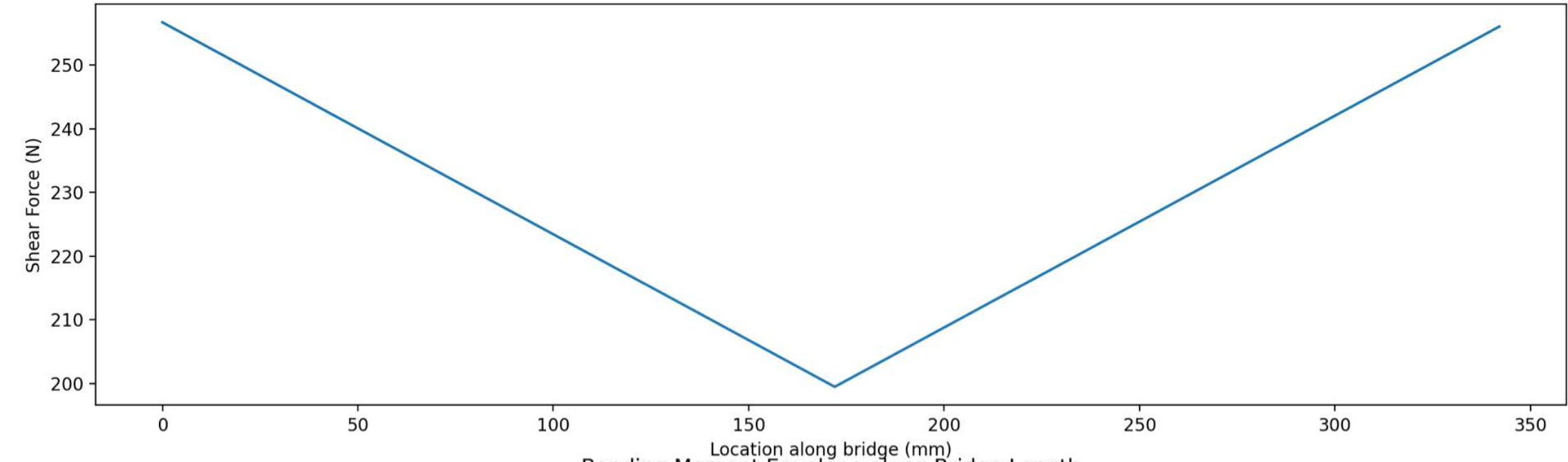
```
----- TESTING FOR DESIGN 0 LOAD CASE 1 -----
Testing at location x_start = 172 mm
I_current = 404154.88612259005 mm^4 y_current = 40.42535607402431 mm
Support A: 199.5 Support B: 199.5
Max Shear Force (N): 199.5
Max Bending Moment (N*mm): 68628.0
Flexural stresses (MPa) at top and bottom: (6.086642294374483, 6.864475556053586)
76.27 1.27 1 40.42535607402431
d_web = 17.287321962987846 d_top = 35.20964392597568 A_web = 87.81959557197825 A_top = 127.0
Q = 5989.790401911081
Shear stress: 1.164053592015846
Shear strength of cement (MPa): 0.11036476038791357
Type 1 buckling stress (MPa): 3.454575724504563
Type 2 buckling stress (MPa): 0
Type 3 buckling stress (MPa): 27.742814477165346
Type 4 buckling stress (MPa): 5.256619850378164
----- SAFETY CHECKS -----
76.27 1.27 1 40.42535607402431
d_web = 17.287321962987846 d_top = 35.20964392597568 A_web = 87.81959557197825 A_top = 127.0
Q = 5989.790401911081
Fail: Top flexural stress 6.09 MPa exceeds compressive strength 6 MPa
Safety facrtor for tension: 4.370326582858009 Safe
Safety facrtor for shear: 3.4362679067662283 Safe
Safety facrtor for shear at glue: 18.12172647292792 Safe
Fail: Type 1 buckling stress 3.45 MPa less than compressive strength 6.086642294374483 MPa
Safety facrtor for Type 2 buckling: 0.0 Safe
Safety facrtor for Type 3 buckling: 4.557983389759302 Safe
Safety facrtor for Type 4 buckling: 4.515788522481195 Safe
Some checks failed: Structure is NOT safe. Lowest factor of safety: 0.57, it will fail if loads are increased by a weight of 227.03 N or more.
Max bending moment at all x (N*mm): 69271.72
Max shear force at all x (N): 256.69
```

Our design:

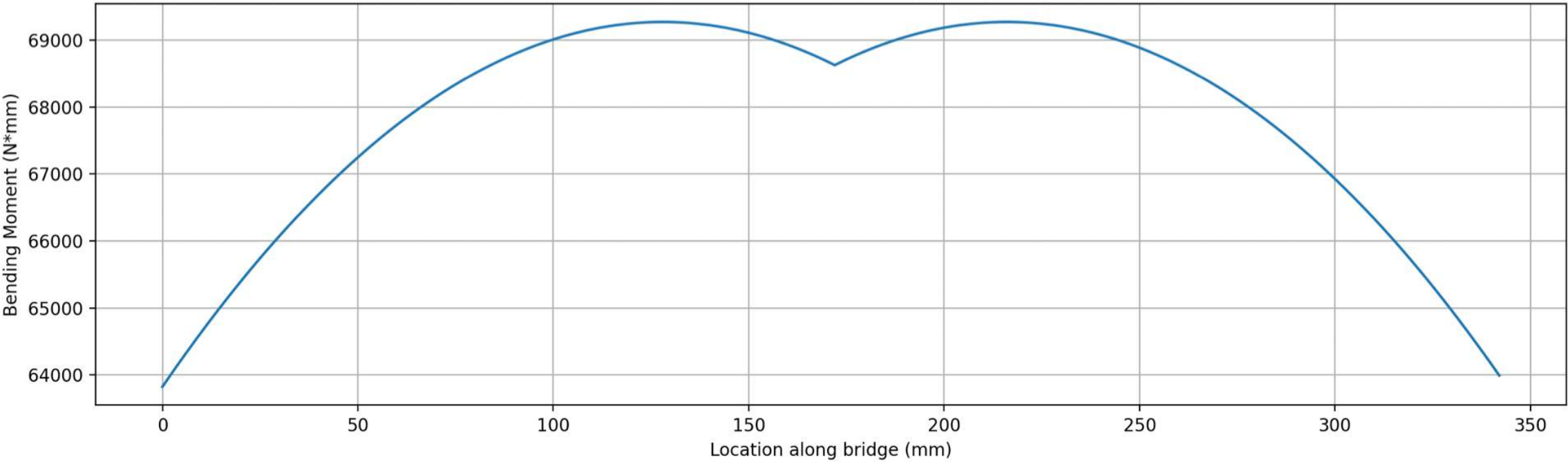
```
----- TESTING FOR BASE LOAD CASE 2 -----
I_current = 952798.7759833706 mm^4 y_current = 62.01286824166984 mm
Setting train loads with first car weight: 135 N, second car weight: 135 N, locomotive weight: 182 N
Location of loads (mm): [172, 348, 512, 688, 852, 1028]
Total moment about A: 287180.0 N*mm Support reaction B calculated as: 239.31666666666666 N
Total loads: 452.0 N
Support A: 212.68333333333334 Support B: 239.31666666666666
Max Shear force: 239.31666666666666
Max Bending Moment (N*mm): 76666.13333333333
Flexural stresses (MPa) at top and bottom: (3.0566018573326446, 4.989812061934544)
100 1.27 2 62.01286824166984
d_web = 17.723565879165076 d_top = 36.71713175833016 A_web = 90.03571466615858 A_top = 254.0
Q = 10921.905386979233
Shear stress: 1.0800315641098968
Shear strength of cement (MPa): 0.11712354896318142
Type 1 buckling stress (MPa): 8.84371385473168
Type 2 buckling stress (MPa): 0
Type 3 buckling stress (MPa): 26.393911176209954
Type 4 buckling stress (MPa): 5.750589269847865
----- SAFETY CHECKS -----
100 1.27 2 62.01286824166984
d_web = 17.723565879165076 d_top = 36.71713175833016 A_web = 90.03571466615858 A_top = 254.0
Q = 10921.905386979233
Safety facrtor for compression: 1.9629641935884719 Safe
Safety facrtor for tension: 6.012250487119355 Safe
Safety facrtor for shear: 3.7035954623201985 Safe
Safety facrtor for shear at glue: 17.075985296763108 Safe
Safety facrtor for Type 1 buckling: 2.893315605863428 Safe
Safety facrtor for Type 2 buckling: 0.0 Safe
Safety facrtor for Type 3 buckling: 8.635050427942454 Safe
Safety facrtor for Type 4 buckling: 5.3244640813689434 Safe
All checks passed: Structure is safe.
```

Design O graphs

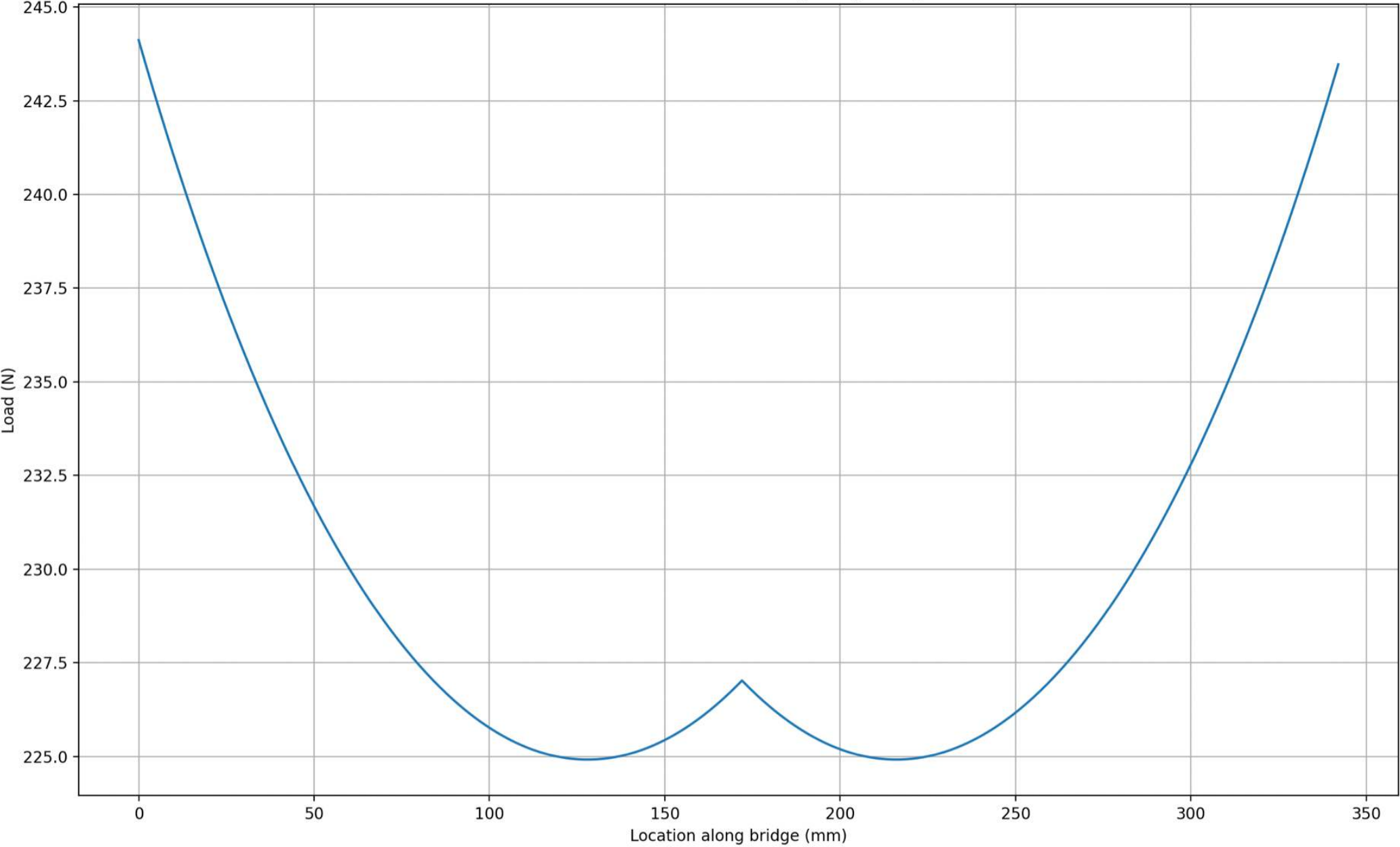
Shear Force Envelope along Bridge Length



Bending Moment Envelope along Bridge Length

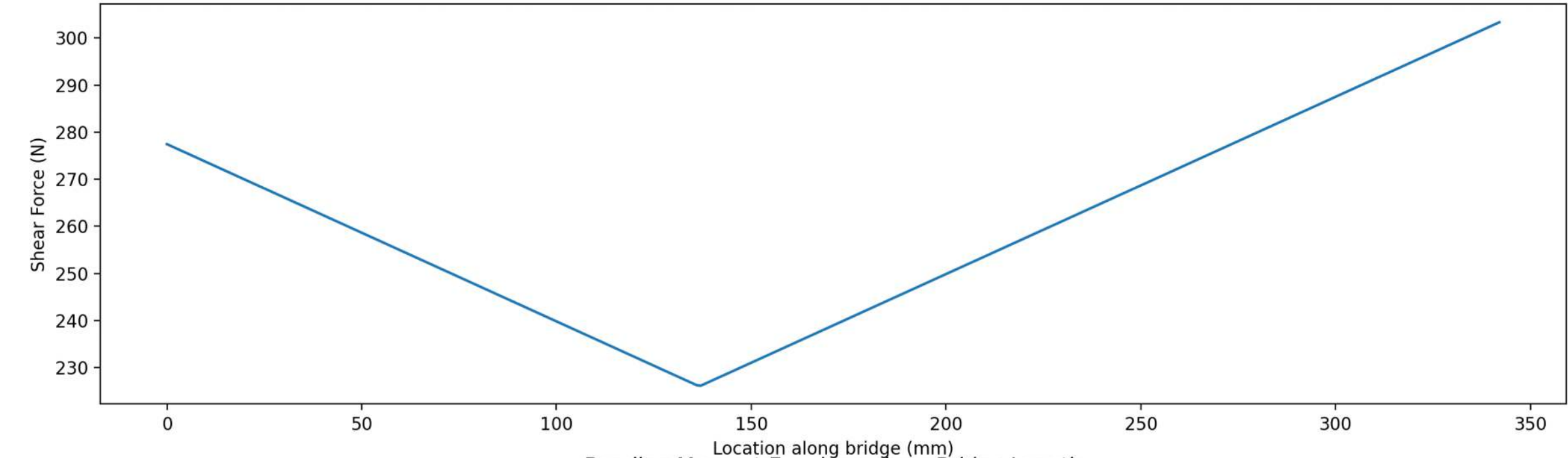


Max load before failure along Bridge Length

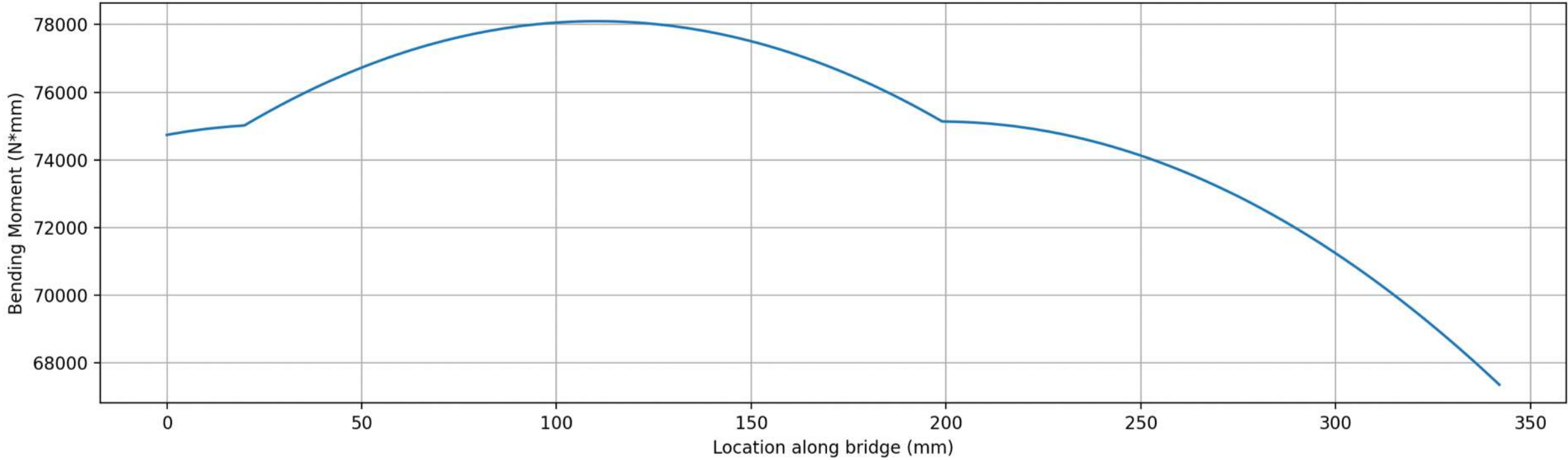


Final design graphs

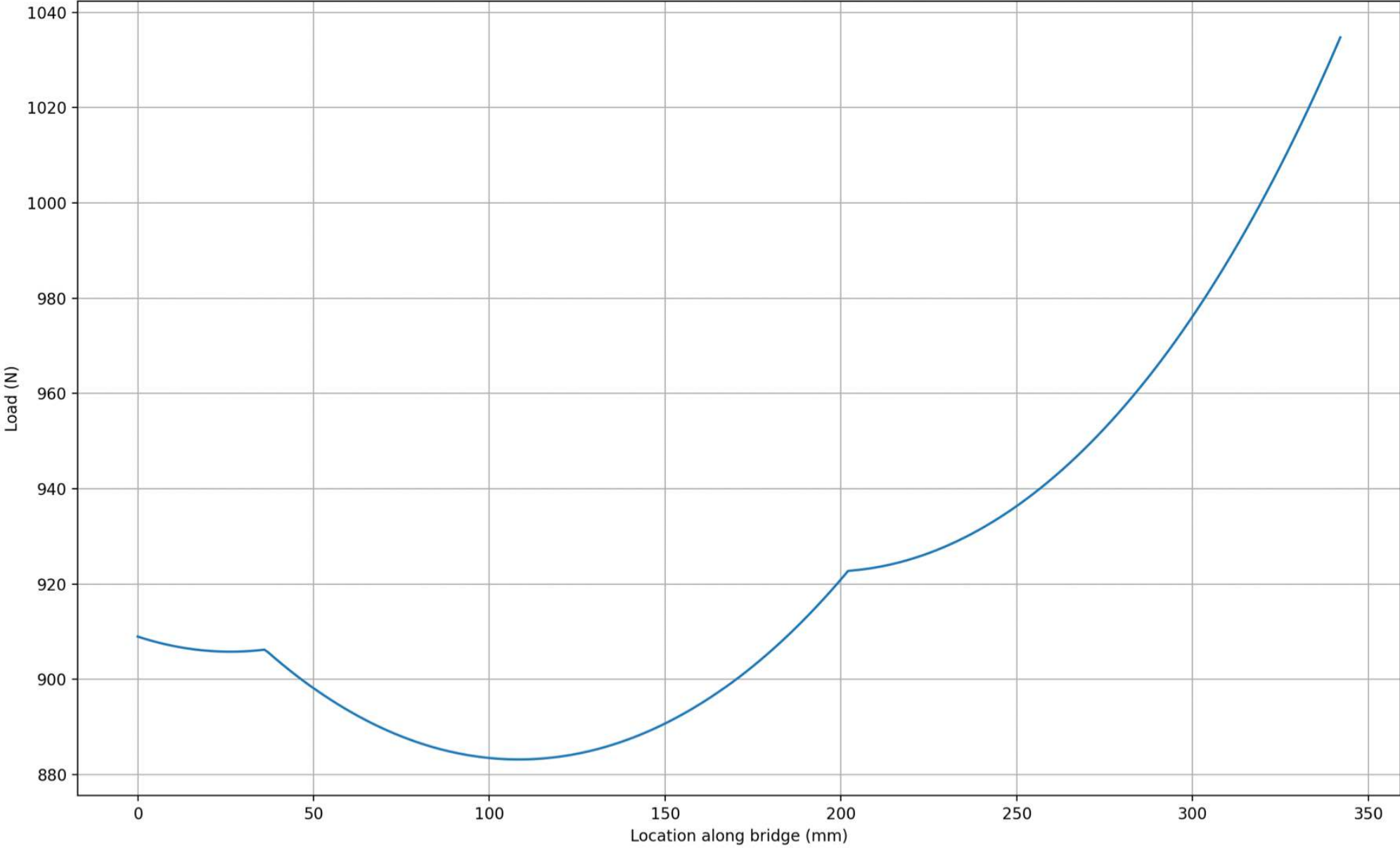
Shear Force Envelope along Bridge Length



Bending Moment Envelope along Bridge Length



Max load before failure along Bridge Length



Link to full code:

<https://github.com/JanKazimierzak/CIV102bridge>

```
import math
def initialize():
    global L
    global t
    global n_of_webs
    global n_plates_top
    global n_plates_bottom
    global b_top
    global b_bot
    global first_car_weight
    global second_car_weight
    global locomotive_weight
    global dist_wheels_car
    global dist_cars_train
    global depth
    global d_web_plates
    global support_reaction_A
    global support_reaction_B
    global I_manual
    global y_centroid_manual
    global b_support
    global E
    global mu
    global b_in
    global b_out
    global a
    global loads
    global positions
    global tensile_strength
    global compressive_strength
    global shear_strength
    global shear_strength_cement
    global I_current
    global y_current
    global x_start
    global glue_tabs

    L = 1250 #mm
    t = 1.27 #mm
    n_of_webs = 2
    n_plates_top = 2
    n_plates_bottom = 1
    b_top = 100 #mm
    b_bot = 80 #mm
    depth = 100 #mm
    first_car_weight = 135 #N
    desired_weight(N):
        global first_car_weight
        global second_car_weight
        global locomotive_weight
        second_car_weight = N/(3.618)
        first_car_weight = second_car_weight * 1.1
        locomotive_weight = first_car_weight * 1.38
        #print("First car weight set to:", first_car_weight, "N")
        #print("Second car weight set to:", second_car_weight, "N")
        #print("Locomotive weight set to:", locomotive_weight, "N")

def type_1():
    global E
    global mu
    global t
    global b_in
    stress = 0
    if b_in != 0:
        stress = ((4*math.pi**2*E)/(12*(1-mu**2)))*(t*n_plates_top/b_in)**2
    return stress

def type_2():
    global E
    global mu
    global t
    global b_out
    stress = 0
    if b_out != 0:
        stress = ((0.425*math.pi**2*E)/(12*(1-mu**2)))*(t*n_plates_top/b_out)**2
    return stress

# Function that will plot all the required figures
def sfe():
    global sfe_shears
    global locations_sfe
    sfe_shears = []
    locations_sfe = []
    bme_moments = []
    factor_safety = []
    for x in range(0, L-908 +1, 1):
        x_start = x
        train_loads(x_start)
        support_reactions()
        sfe_shears.append(SFD_diagram(L, loads, positions, support_reaction_A, support_reaction_B))
        bme_moments.append(max_BMD(L, loads, positions, support_reaction_A, support_reaction_B))
        locations_sfe.append(x)
        factor_safety.append(check_safety())

    plt.figure(1)
    plt.subplot(211)
    plt.plot(locations_sfe, sfe_shears)
    plt.ylabel("Shear Force (N)")
    plt.title("Shear Force Envelope along Bridge Length")
    plt.xlabel("Location along bridge (mm)")
    plt.subplot(212)
    plt.plot(locations_sfe, bme_moments)
    plt.xlabel("Location along bridge (mm)")
    plt.ylabel("Bending Moment (N*mm)")
    plt.title("Bending Moment Envelope along Bridge Length")
    plt.grid()
    plt.show()

    plt.figure(2)
    plt.plot(locations_sfe, factor_safety)
    plt.xlabel("Location along bridge (mm)")
    plt.ylabel("Load (N)")
    plt.title("Max load before failure along Bridge Length")
    plt.grid()
    plt.show()

def support_reactions():
    global support_reaction_A
    global support_reaction_B
    total_moment = 0.0
    for i in range(len(loads)):
        total_moment += loads[i] * (positions[i])
    support_reaction_B = total_moment / 1200
    #print("Total moment about A:", total_moment, "N*mm", "Support reaction B calculated as:", support_reaction_B, "N")
    #print("Total loads:", sum(loads), "N")
    support_reaction_A = sum(loads) - support_reaction_B

second_car_weight = 135 #N
locomotive_weight = 182 #N
dist_wheels_car = 176 #mm
dist_cars_train = 164 #mm
d_web_plates = 25 #mm
support_reaction_A = 0
support_reaction_B = 0
I_manual = 85e3 #mm^4
y_centroid_manual = 42.51 #mm
b_support = 50 #mm
E = 4000 #MPa N/mm^2
mu = 0.2
b_in = 100 #mm
b_out = 0 #mm
a=100 #mm
tensile_strength = 30 #MPa
compressive_strength = 6 #MPa
shear_strength = 4 #MPa
shear_strength_cement = 2 #MPa
I_current = 0
y_current = 0
x_start = 172 #mm Center at 172 mm, min at 0 mm, max at L-908 mm
glue_tabs = 10 #mm

def shear_strength_cement_calc():
    global shear_strength_cement
    Q = n_plates_top * b_top * t * (depth - n_plates_top*t/2 - y_current)
    #print("Q in shear strength of cement calc =", Q)
    shear_at_glue = (max_sfd*Q)/(n_of_webs*glue_tabs*I_current)
    return shear_at_glue
    #print("Shear strength of cement set to:", shear_strength_cement, "MPa")

def max_bending_all_x():
    max_bending = 0.0
    for x in range(0, L-908 +1, 1):
        x_start = x
        train_loads(x_start)
        support_reactions()
        if max_BMD(L, loads, positions, support_reaction_A, support_reaction_B) > max_bending:
            max_bending = max_BMD(L, loads, positions, support_reaction_A, support_reaction_B)
    return max_bending

def type_3():
    global E
    global mu
    global t
    distance = depth - n_plates_top*t - y_current
    stress = ((6*math.pi**2*E)/(12*(1-mu**2)))*(t/distance)**2
    return stress

def type_4():
    global E
    global mu
    global t
    global b_in
    global a
    stress = ((5*math.pi**2*E)/(12*(1-mu**2)))*((t/(depth-n_plates_top*t-n_plates_bottom*t))**2+(t/a)**2)
    return stress

def flexural_stress(I):
    # ensure loads/positions/supp reactions are set
    global n_of_webs, n_plates_top, n_plates_bottom, t, b_top, b_bot, depth
    # get centroid and section I if caller didn't pass an I

    # get signed bending moment with largest magnitude from max_BMD
    max_abs_M = max_BMD(L, loads, positions, support_reaction_A, support_reaction_B)
    # keep sign of the largest magnitude moment
    M = max_abs_M

    # distances to extreme fibers from centroid (mm)
    y_bot = float(y_current)
    y_top = float(depth) - float(y_current)

    # bending stress = M * y / I (N/mm^2 = MPa)
    sigma_top = M * y_top / I
    sigma_bot = M * y_bot / I

    # return (sigma_top_MPa, sigma_bot_MPa)
    return sigma_top, sigma_bot

def Q_calculation():
    A_top = b_top * n_plates_top*t
    A_web = n_of_webs*t * (depth - t*(n_plates_top) - y_current)
    d_web = (depth - t*n_plates_top - y_current)/2
    d_top = (depth - n_plates_top*t/2 - y_current)
    print(depth, t, n_plates_top, y_current)
    print("d_web =", d_web, "d_top =", d_top, "A_web =", A_web, "A_top =", A_top)
    Q = A_top * d_top + A_web * d_web
    print("Q =", Q)
    return Q

def x_for_mid_location(L):
    train_len = 3*dist_wheels_car + 2*dist_cars_train
    return (L-train_len)/2 - b_support/2

def shear_stress(I):
    # ensure loads/positions/supp reactions are set
    global n_of_webs, n_plates_top, n_plates_bottom, t, b_top, b_bot, depth
    # get centroid and section I if caller didn't pass an I

    # get shear force with largest magnitude from SFD
    max_shear = max_sfd

    # Q calculation
    Q = Q_calculation()

    # shear stress = V * Q / (I * t) (N/mm^2 = MPa)
    tau_max = max_shear * Q / (I * t*n_of_webs)

    return tau_max

def train_loads(x_start):
    global loads, positions, first_car_weight, second_car_weight, locomotive_weight
    #print("Setting train loads with first car weight:", first_car_weight, "N, second car weight:", second_car_weight, "N")
    loads = [first_car_weight/2, first_car_weight/2, second_car_weight/2, second_car_weight/2, locomotive_weight/2, locomotive_weight/2]
    positions = [x_start, x_start + dist_wheels_car, x_start + dist_wheels_car + dist_cars_train, x_start + 2*dist_wheels_car + dist_cars_train, x_start + 3*dist_wheels_car + dist_cars_train, x_start + 3*dist_wheels_car + dist_cars_train + dist_wheels_car]
    # Locomotive
    #print("Location of loads (mm):", positions)
```

```

def I_calculation(n_of_webs, n_plates_top, n_plates_bottom, t, b_top, b_bot, depth):
    # web free height (remove both top and bottom plate thicknesses)
    h_web = float(depth) - (n_plates_top + n_plates_bottom) * float(t)
    #print("h_web =", h_web)
    if h_web <= 0:
        raise ValueError("Invalid geometry: web height <= 0")

    # areas (mm^2)
    A_web = float(n_of_webs) * float(t) * h_web
    A_top = float(n_plates_top) * float(b_top) * float(t)
    A_bot = float(n_plates_bottom) * float(b_bot) * float(t)
    #print("A_web =", A_web, "A_top =", A_top, "A_bot =", A_bot)
    A_total = A_web + A_top + A_bot
    if A_total == 0.0:
        raise ValueError("Total area is zero")

    # y positions from bottom datum (mm)
    y_bot = n_plates_bottom*float(t) / 2.0
    # web centroid sits above the bottom plates
    y_web = (float(n_plates_bottom) * float(t) + h_web/2 )
    y_top = depth - (n_plates_top*float(t) / 2.0)
    #print("y_bot =", y_bot, "y_web =", y_web, "y_top =", y_top)
    # centroid (mm)
    y_centroid = (A_bot * y_bot + A_web * y_web + A_top * y_top) / A_total
    #print("y_centroid =", y_centroid)
    # individual second moments about their own centroids (mm^4)
    I_web_ind = (n_of_webs*float(t) * h_web**3) / 12.0
    I_top_ind = (float(b_top) * (n_plates_top*float(t))**3) / 12.0
    I_bot_ind = (float(b_bot) * (n_plates_bottom*float(t))**3) / 12.0
    #print("I_web_ind =", I_web_ind, "I_top_ind =", I_top_ind, "I_bot_ind =", I_bot_ind)
    # total I using parallel axis theorem
    #print(y_top - y_centroid)
    I_web = (I_web_ind + (A_web* (y_centroid - y_web)**2))
    I_top = (I_top_ind + ((A_top) * (y_top - y_centroid)**2))
    I_bot = (I_bot_ind + (A_bot) * (y_centroid - y_bot)**2)
    #print("I_web =", I_web, "I_top =", I_top, "I_bot =", I_bot)

    I_total = I_web + I_top + I_bot

    return y_centroid, I_total

def print_geometry():
    print("Geometry:")
    print(f"  n_of_webs = {n_of_webs}")
    print(f"  n_plates_top = {n_plates_top}")
    print(f"  n_plates_bottom = {n_plates_bottom}")
    print(f"  t = {t} mm")
    print(f"  b_top = {b_top} mm")
    print(f"  b_bot = {b_bot} mm")
    print(f"  depth = {depth} mm")

def results():
    train_loads(x_start)
    support_reactions()
    print("Support A:", support_reaction_A, "Support B:", support_reaction_B)
    SFD_diagram(L, loads, positions, support_reaction_A, support_reaction_B)
    print("Max Shear Force (N):", max_sfd)
    max_BMD(L, loads, positions, support_reaction_A, support_reaction_B)
    print("Max Bending Moment (N*mm):", max_BMD(L, loads, positions, support_reaction_A, support_reaction_B))
    print("Flexural stresses (MPa) at top and bottom:", flexural_stress(I_current))
    #print("Q =", Q_calculation())
    print("Shear stress:", shear_stress(I_current))
    print("Shear strength of cement (MPa):", shear_strength_cement_calc())
    print("Type 1 buckling stress (MPa):", type_1())
    print("Type 2 buckling stress (MPa):", type_2())
    print("Type 3 buckling stress (MPa):", type_3())
    print("Type 4 buckling stress (MPa):", type_4())

```

```

def SFD_diagram(L, loads, positions, support_reaction_A, support_reaction_B):
    global max_sfd

    shear = support_reaction_A
    max_sfd = abs(shear)

    for load in loads:
        shear = shear - load
        if abs(shear) > max_sfd:
            max_sfd = abs(shear)

    shear = shear + support_reaction_B
    if abs(shear) > max_sfd:
        max_sfd = abs(shear)
    return max_sfd

def max_BMD(L, loads, positions, support_reaction_A, support_reaction_B):
    global max_bending_moment

    shear = support_reaction_A
    moment = 0
    max_bending_moment = 0

    for i in range(len(loads)):
        load = loads[i]
        pos = positions[i]
        dist = pos - (0 if i == 0 else positions[i-1])
        moment += shear * dist
        if abs(moment) > abs(max_bending_moment):
            max_bending_moment = moment
        shear -= load

    last_dist = L - positions[-1]
    moment += shear * last_dist
    if abs(moment) > abs(max_bending_moment):
        max_bending_moment = moment

    return abs(max_bending_moment)

def check_safety():
    sigma_top, sigma_bot = flexural_stress(I_current)
    tau_max = shear_stress(I_current)
    fail_factor_of_safety = 100
    safe = True
    if abs(sigma_top) > compressive_strength:
        print(f"\033[91mFail: Top flexural stress {sigma_top:.2f} MPa exceeds compressive strength {compressive_strength} MPa\033[0m")
        safe = False
        if compressive_strength/abs(sigma_top) < fail_factor_of_safety:
            fail_factor_of_safety = compressive_strength/abs(sigma_top)
    else:
        print("Safety facrtor for compression:", compressive_strength/abs(sigma_top), "\033[92mSafe \033[91m\033[0m")
    if abs(sigma_bot) > tensile_strength:
        print(f"\033[91mFail: Bottom flexural stress {sigma_bot:.2f} MPa exceeds tensile strength {tensile_strength} MPa\033[0m")
        safe = False
        if tensile_strength/abs(sigma_bot) < fail_factor_of_safety:
            fail_factor_of_safety = tensile_strength/abs(sigma_bot)
    else:
        print("Safety facrtor for tension:", tensile_strength/abs(sigma_bot), "\033[92mSafe \033[91m\033[0m")
    if abs(tau_max) > shear_strength:
        print(f"\033[91mFail: Shear stress {tau_max:.2f} MPa exceeds shear strength {shear_strength} MPa\033[0m")
        safe = False
        if shear_strength/abs(tau_max) < fail_factor_of_safety:
            fail_factor_of_safety = shear_strength/abs(tau_max)
    else:
        print("Safety facrtor for shear:", shear_strength/abs(tau_max), "\033[92mSafe \033[91m\033[0m")
    if shear_strength_cement_calc() > shear_strength_cement:
        print(f"\033[91mFail: Shear at glue {shear_strength_cement_calc():.2f} MPa exceeds shear strength of cement {shear_strength_cement} MPa\033[0m")
        safe = False
        if shear_strength_cement_calc()/shear_strength_cement < fail_factor_of_safety:
            fail_factor_of_safety = shear_strength_cement_calc()/shear_strength_cement
    else:
        print("Safety facrtor for shear at glue:", shear_strength_cement/shear_strength_cement_calc(), "\033[92mSafe \033[91m\033[0m")
    if type_1() < sigma_top and type_1() != 0:
        print(f"\033[91mFail: Type 1 buckling stress {type_1():.2f} MPa less than compressive strength {sigma_top} MPa\033[0m")
        safe = False
        if type_1()/sigma_top < fail_factor_of_safety:
            fail_factor_of_safety = type_1()/sigma_top
    else:
        print("Safety facrtor for Type 1 buckling:", type_1()/sigma_top, "\033[92mSafe \033[91m\033[0m")
    if type_2() < sigma_top and type_2() != 0:
        print(f"\033[91mFail: Type 2 buckling stress {type_2():.2f} MPa less than compressive strength {sigma_top} MPa\033[0m")
        safe = False
        if type_2()/sigma_top < fail_factor_of_safety:
            fail_factor_of_safety = type_2()/sigma_top
    if type_3() < sigma_top:
        print(f"\033[91mFail: Type 3 buckling stress {type_3():.2f} MPa less than compressive strength {sigma_top} MPa\033[0m")
        safe = False
        if type_3()/sigma_top < fail_factor_of_safety:
            fail_factor_of_safety = type_3()/sigma_top
    else:
        print(f"\033[92mSafe \033[91m\033[0m")
    if type_4() < tau_max:
        print(f"\033[91mFail: Type 4 buckling stress {type_4():.2f} MPa less than compressive strength {tau_max} MPa\033[0m")
        safe = False
        if type_4()/tau_max < fail_factor_of_safety:
            fail_factor_of_safety = type_4()/tau_max
    else:
        print("Safety facrtor for Type 4 buckling:", type_4()/tau_max, "\033[92mSafe \033[91m\033[0m")
    if safe:
        print("\033[92mAll checks passed: Structure is safe.\033[0m")
    if not safe:
        print(f"\033[91mSome checks failed: Structure is NOT safe. Lowest factor of safety: {fail_factor_of_safety:.2f}, it will fail if loads are inc")

```

```
442 print("\033[92mSafe \033[91mFail\033[0m")
443 #print("First car weight:", first_car_weight, "N", "Second car weight:", second_car_weight, "N", "Locomotive weight:", locomotive_weight,"N")
444 print("\033[93m ----- TESTING FOR BASE LOAD CASE 2 -----\033[00m")
445 initialize()
446 print("Testing at location x_start =", x_start, "mm")
447 #desired_weight(2000)
448 I_current = I_calculation(n_of_webs, n_plates_top, n_plates_bottom, t, b_top, b_bot, depth)[1]
449 y_current = I_calculation(n_of_webs, n_plates_top, n_plates_bottom, t, b_top, b_bot, depth)[0]
450 print("I_current =", I_current, "mm^4", "y_current =", y_current, "mm")
451 results()
452 print("\033[93m ----- SAFETY CHECKS -----\033[00m")
453 check_safety()
454 print("Max bending moment at all x (N*mm):", max_bending_all_x())
455 print("Max shear force at all x (N):", max_shear_all_x())
456 #print(I_calculation(n_of_webs, n_plates_top, n_plates_bottom, t, b_top, b_bot, depth))
457 #print(I_manual, y_centroid_manual)
458 # y, I = I_calculation(n_of_webs, n_plates_top, n_plates_bottom, t, b_top, b_bot, depth)
459 # print(f"y_centroid = {y:.6f} mm")
460 # print(f"I_total = {I:.6f} mm^4  ({I/1e4:.6f} x10^4 mm^4)")
461 # train_loads(172)
462 # support_reactions()
463 # print(x_for_mid_location(L))
464 # print("Support A:", support_reaction_A, "Support B:", support_reaction_B)
465 # SFD_diagram(L, loads, positions, support_reaction_A, support_reaction_B)
466 # max_BMD(L, loads, positions, support_reaction_A, support_reaction_B)
467 # print("Flexural stresses (MPa) at top and bottom:", flexural_stress(I_manual))
468 # print("Q =", Q_calculation())
469 # print("Shear stress:", shear_stress(I_manual))
```